Linear Least Squares

INFO 521 · Module 1 — *reference deck*

Greg Chism
School of Information · University of Arizona

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Learning outcomes

By the end of this module you should be able to:

1. **Placeholder outcome** — state the linear model and the squared-error loss $\mathcal{L}(\mathbf{w})$ in PML notation.
2. **Placeholder outcome** — derive the normal equations and solve for $\hat{\mathbf{w}}$.
3. **Placeholder outcome** — fit a line to real data and read the result critically (units, leverage, what the slope means).

The least-squares problem

A section divider — set up the geometry before the algebra.

The normal equations

We model targets as a linear function of the inputs and minimize squared error:

$$\mathcal{L}(\mathbf{w}) \;=\; \frac{1}{N}\,\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2.$$

Setting $\nabla_{\mathbf{w}}\mathcal{L} = \mathbf{0}$ gives the **normal equations** $\mathbf{X}^\top\mathbf{X}\,\mathbf{w} = \mathbf{X}^\top\mathbf{y}$, so when $\mathbf{X}^\top\mathbf{X}$ is invertible,

$$\boxed{\hat{\mathbf{w}} = \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\mathbf{y}}$$

A least-squares fit to NHANES

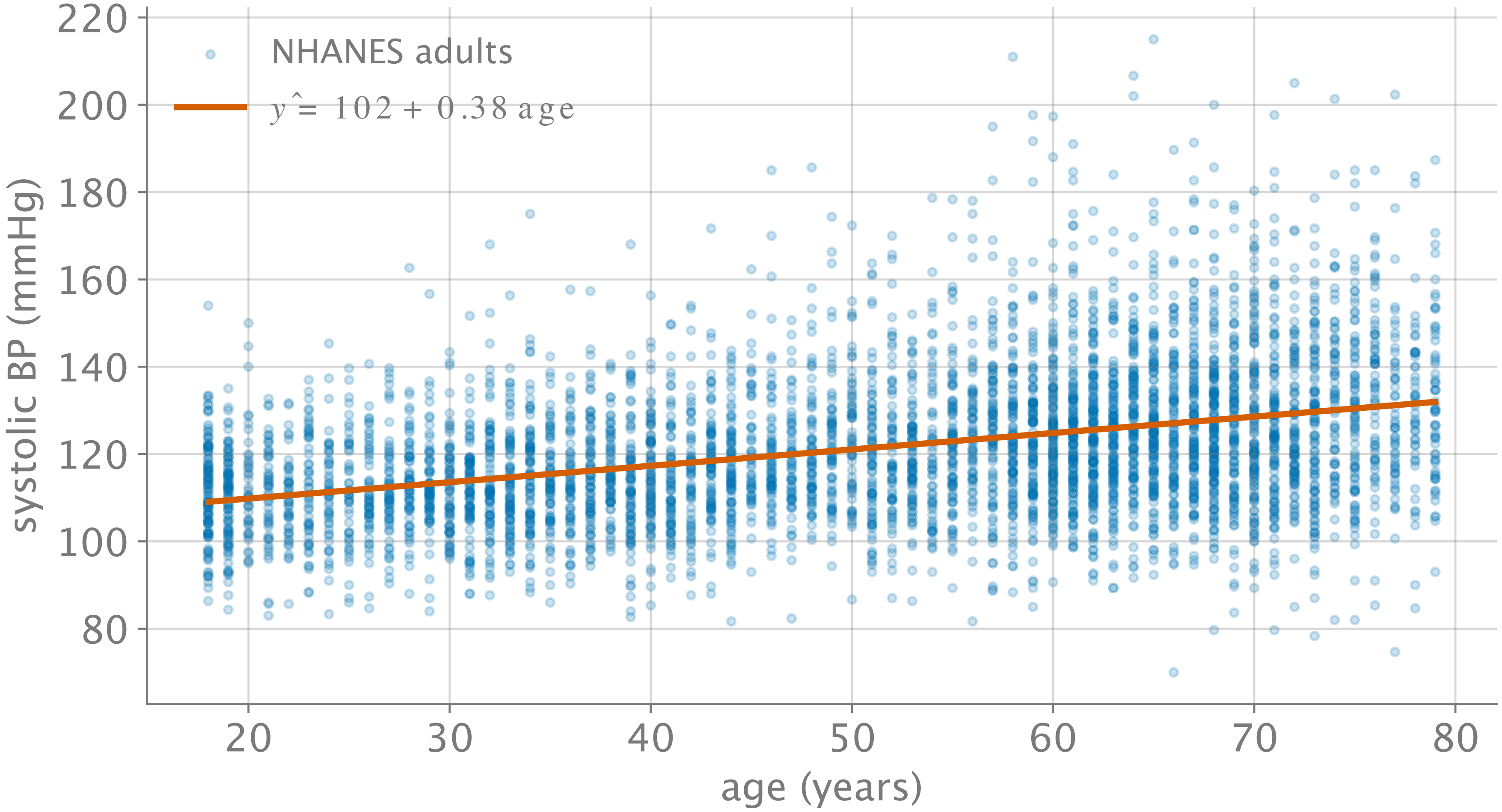

Figure 1: Systolic BP vs. age, NHANES 2021–2022, with the least-squares line.

Live demo: least-squares explorer

Example Domain

This domain is for use in documentation examples without needing permission.
Avoid use in operations.

Learn more

Explorer — static fallback

The iframe above does not print or export to PDF; this snapshot stands in.

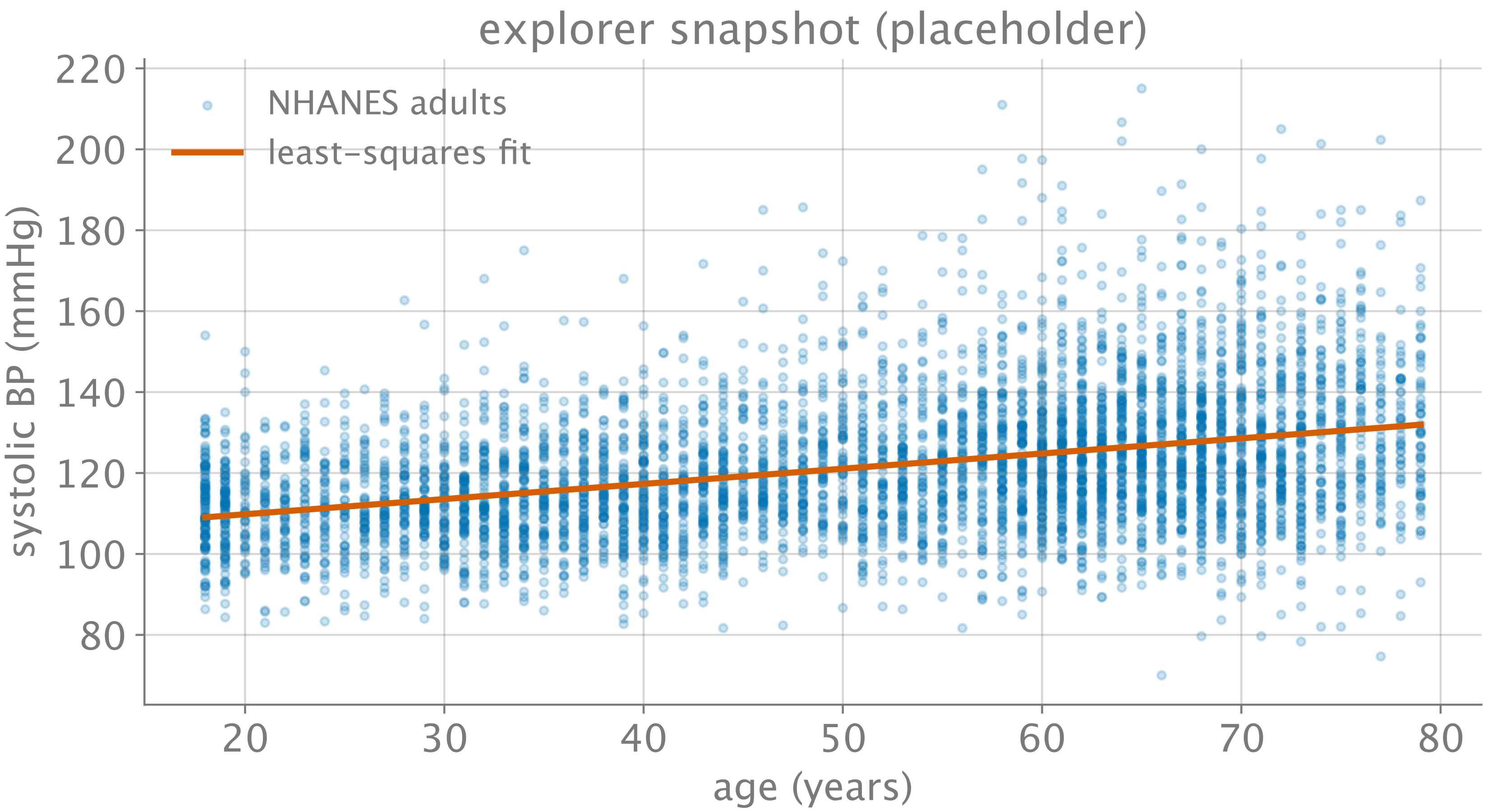

Figure 2: Static fallback snapshot of the explorer (used in print / PDF).

Checkpoint preview

Next, in the peer-engagement loop you will:

- **Placeholder** — fit the line yourself in NumPy and report $\hat{\mathbf{w}}$.
- **Placeholder** — interpret the slope in clinical units (mmHg per year).
- **Placeholder** — reflect, peer-review, and revise toward the mastery token.